

Notes: Scharfstein & Stein (JF, 2000)

The Dark Side of Internal Capital Markets: Divisional Rent-Seeking and Inefficient Investment

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to standard agency theories of diversification	3
2.2	Contribution relative to efficient internal capital market theories	3
2.3	Contribution relative to rent-seeking and influence-cost models	4
2.4	Contribution relative to empirical conglomerate-discount studies	4
3	Model primitives and environment	4
3.1	Agents and organizational structure	4
3.2	Production at time 1	4
3.3	Production at time 2	5
3.4	Rent-seeking technology	5
3.5	Efficient benchmark allocation	5
4	Cash-only benchmark	5
4.1	Retention constraint	5
4.2	Rent-seeking condition	6
4.3	Main cash-only implication	6
5	CEO-level agency and capital-budget compensation	6
5.1	CEO's contracting environment	6
5.2	Operating budget and agency problem	6
5.3	Retention constraint with flexible capital allocation	6
5.4	CEO objective	7
5.5	Low, medium, and high productivity divisions	7
6	Main theoretical results	7
6.1	Proposition 1: regions of rent-seeking and capital allocation	7
6.2	Why the strongest division can subsidize the weakest division	7
6.3	Why distortions disappear when both divisions are weak	8
6.4	Proposition 2: magnitude of the investment distortion	8
7	Investor choice of capital and operating budgets	8
7.1	Operating budget W	8
7.2	Capital budget I	8
8	Extensions and variations	8
8.1	Signal-jamming interpretation	8
8.2	Breakups	9
8.3	Rent-seeking as signal-jamming or organizational politics	9

9	Empirical implications	9
9.1	Prediction 1: weak divisions are subsidized when paired with strong divisions	9
9.2	Prediction 2: misallocation is strongest with high divisional imbalance	9
9.3	Prediction 3: CEOs with weaker incentives tolerate larger distortions	9
9.4	Prediction 4: breakups and refocusing reduce distortions	9
10	Tables, figures, and original equation panels	10
10.1	Panel 1: model primitives and rent-seeking	10
10.2	Panel 2: cash-only benchmark	10
10.3	Panel 3: low, medium, and high productivity thresholds	11
10.4	Panel 4: Proposition 1	12
10.5	Panel 5: Proposition 2	12
10.6	Panel 6: signal-jamming extension	13
10.7	Panel 7: empirical implications	13
11	Short summary	14
12	Conclusion	15
12.1	Core conclusion	15
12.2	Economic intuition	15
12.3	Incremental contribution takeaway	15
12.4	Final takeaway	15

1 Big picture

- **Question.** Why might internal capital markets inside conglomerates allocate capital inefficiently, even when a headquarters CEO has formal authority to move funds across divisions?
- **Main answer.** Divisional managers can engage in rent-seeking to raise their bargaining power. Because the CEO is also an agent of outside investors, she may find it privately cheaper to compensate a rent-seeking division manager with extra capital rather than with cash wages. This creates inefficient cross-subsidization.
- **Core mechanism.** Weak divisions have managers with lower opportunity costs of rent-seeking. When a weak division is paired with a strong division, the weak manager is more likely to rent-seek, and the CEO can satisfy the weak manager by tilting investment toward that weak division.
- **Why this is a “dark side” of internal capital markets.** A common argument for internal capital markets is that headquarters can reallocate resources to high-return divisions. This paper shows that headquarters may instead move capital toward politically powerful or rent-seeking divisions.
- **Two-tiered agency problem.** The agency problem is not only between outside investors and the CEO. It is also between the CEO and division managers. The interaction of these two agency layers generates distortions.
- **Socialism prediction.** The model predicts a kind of internal socialism: weak divisions receive subsidies from strong divisions, especially when productivity differences across divisions are large.
- **Key theoretical result.** Investment distortions are most likely when one division is low productivity and the other division is high productivity. In that case, only the weak-division manager rent-seeks, and headquarters tilts capital toward the weak division.
- **Magnitude result.** Distortions can be large when the CEO has low-powered incentives to maximize firm value. A benevolent principal might tilt capital slightly, but an agency-prone CEO can tolerate much larger marginal-product gaps across divisions.
- **Extensions.** The model can be reinterpreted through signal-jamming incentives, breakups, or budgeting practices. The key requirement is that the CEO cares about conserving cash wages and has discretion over divisional capital allocation.
- **Empirical implications.** The theory predicts that conglomerates overinvest in weaker divisions, particularly when they are paired with much stronger divisions; breakup pressure should be strongest when distortions are large; and headquarters power may produce internal tug-of-war behavior among divisions.

2 Incremental contribution of the paper

2.1 Contribution relative to standard agency theories of diversification

Earlier agency theories often emphasize that diversified firms may overinvest because CEOs enjoy empire building or private benefits from firm size. Those theories predict excessive aggregate investment but do not automatically explain *misallocation across divisions*. Scharfstein and Stein add a divisional layer to the agency problem. The CEO may want a valuable empire, but she must bargain with division managers whose rent-seeking affects compensation and capital allocation.

Interpretation: The incremental contribution is to show that value loss in conglomerates can come not only from overinvestment, but also from internal capital reallocation that favors low-productivity divisions.

2.2 Contribution relative to efficient internal capital market theories

Williamson-style and Stein-style theories argue that internal capital markets can improve allocation because headquarters can pick winners and reallocate funds toward high-return projects. Scharfstein and Stein show

that the same authority can be corrupted by organizational bargaining. Headquarters may reallocate capital not because it is efficient, but because capital is a noncash way to compensate politically costly division managers.

Interpretation: The paper does not deny that internal capital markets can be useful. It identifies a countervailing force: divisional rent-seeking can make headquarters allocations worse than external capital allocation.

2.3 Contribution relative to rent-seeking and influence-cost models

The paper formalizes how rent-seeking affects capital budgeting in a multi-division firm. Division managers do not simply waste effort lobbying. Their lobbying changes bargaining power and therefore changes the form of compensation. Because the CEO cannot contract directly with outside investors over all internal allocations, the manager's compensation can be hidden inside capital budgets.

Interpretation: The paper connects rent-seeking, bargaining, and internal resource allocation in one model. The distortion arises because the capital budget can serve as implicit compensation.

2.4 Contribution relative to empirical conglomerate-discount studies

Empirical work documents that diversified firms often trade at discounts and may cross-subsidize. This paper provides a microfoundation for a particular pattern of cross-subsidization: weak divisions are subsidized by stronger divisions when paired inside the same firm. This generates sharper empirical predictions than a generic conglomerate-discount story.

Interpretation: The paper's model predicts not just that conglomerates may be inefficient, but when and in what direction capital will be misallocated.

3 Model primitives and environment

3.1 Agents and organizational structure

The firm has two divisions and three kinds of agents:

- **outside investors**, who provide financing and receive verifiable cash flows;
- a **CEO**, who allocates investment across divisions and bargains with division managers;
- two **division managers**, one for each division, who operate divisional assets and may engage in rent-seeking.

Interpretation: The delegation of capital allocation to the CEO is the defining feature of integration. Investors can specify total capital investment, but they cannot perfectly contract on the split across divisions.

3.2 Production at time 1

At time 1, division i produces verifiable output

$$\theta_i f(e_i), \tag{1}$$

where θ_i measures divisional productivity and e_i is productive managerial effort. The function $f(\cdot)$ is increasing and concave.

Division manager i also receives private benefits

$$\gamma \theta_i f(e_i), \tag{2}$$

while the CEO receives private benefits from each division equal to

$$\phi\theta_i f(e_i). \quad (3)$$

Interpretation: Both the division manager and the CEO care about output because output generates private benefits as well as verifiable cash flows.

3.3 Production at time 2

At time 2, the old assets depreciate and new capital is invested. If the original division manager remains, output in division i is

$$\theta_i k(I_i), \quad (4)$$

where I_i is new investment and $k(\cdot)$ is increasing and concave with $k'(0) = \infty$.

If the original manager quits, output falls because the old manager has acquired specific human capital.

Interpretation: This gives the division manager bargaining power at time 2. The CEO wants to retain the manager because replacing him lowers output.

3.4 Rent-seeking technology

A division manager allocates total time h between productive effort e_i and rent-seeking effort r_i :

$$e_i + r_i = h. \quad (5)$$

The baseline rent-seeking interpretation is “resume-polishing.” Rent-seeking increases the manager’s outside option:

$$OutsideOption_i = g(r_i), \quad (6)$$

where $g(\cdot)$ is increasing and concave, with $g(0) = 0$.

Interpretation: Rent-seeking is socially wasteful because it reduces productive effort at time 1, but it may privately benefit the division manager by improving his bargaining position at time 2.

3.5 Efficient benchmark allocation

If investment is allocated efficiently across divisions, the marginal products are equalized:

$$\theta_1 k'(I_1^*) = \theta_2 k'(I_2^*). \quad (7)$$

Interpretation: This is the benchmark against which the model measures distorted capital allocation. If the weak division receives too much capital, its marginal product is lower than that of the strong division.

4 Cash-only benchmark

4.1 Retention constraint

If the CEO must retain manager i using cash compensation only, the retention constraint is

$$w_i + \gamma\theta_i k(I_i^*) \geq g(r_i). \quad (8)$$

The left side is the manager’s total utility from staying: cash wage plus private benefits from running the division at the efficient capital allocation. The right side is the outside option created by rent-seeking.

Interpretation: A division manager with lower θ_i receives fewer time-2 private benefits from staying. Therefore, low-productivity managers are more likely to require cash compensation.

4.2 Rent-seeking condition

If a manager rent-seeks at level r_i^* , the rent-seeking condition is

$$g(r_i^*) > \gamma\theta_i [k(I_i^*) + f(h) - f(h - r_i^*)]. \quad (9)$$

The optimal rent-seeking level satisfies the first-order condition

$$g'(r_i^*) = \gamma\theta_i f'(h - r_i^*). \quad (10)$$

Interpretation: Low-productivity managers are more likely to rent-seek because the private-benefit loss from reducing productive effort is lower when θ_i is low.

4.3 Main cash-only implication

The cash-only benchmark shows that weak divisions have managers with stronger incentives to rent-seek and stronger wage demands. In a cash-only world, the CEO simply pays them wages. There is no capital distortion because the allocation is fixed at I_i^* .

Interpretation: This benchmark sets up the central question: once the CEO can choose capital allocations, will she pay rent-seeking managers with cash or with capital?

5 CEO-level agency and capital-budget compensation

5.1 CEO's contracting environment

The CEO has no wealth. Outside investors finance time-2 investment. The financial contract has two important features:

- investors can specify the total amount of investment that must be converted into physical capital;
- investors cannot specify how the total capital budget is divided across divisions.

The CEO also controls the hiring and compensation of division managers.

Interpretation: Capital is protected at the firm-wide level, but divisional allocation is not verifiable. This is the control right that allows the CEO to use capital allocation as implicit compensation.

5.2 Operating budget and agency problem

Outside investors may give the CEO an operating budget W to pay division-manager wages. But if the CEO does not spend this budget on wages, she can use it for herself or other private-benefit activities. Therefore, she prefers to conserve cash wages.

Interpretation: This is the key CEO-level agency friction. Cash wages are costly to the CEO, while distorted capital allocations are only partially costly because the CEO internalizes only a fraction ϕ of output value.

5.3 Retention constraint with flexible capital allocation

When investment can differ from the efficient benchmark, manager i 's retention constraint becomes

$$w_i + \gamma\theta_i k(I_i) \geq g(r_i). \quad (11)$$

Interpretation: The CEO can satisfy this constraint in two ways: by paying a cash wage w_i or by increasing I_i so that the division manager receives more private benefits from capital.

5.4 CEO objective

The CEO chooses investment and wages to maximize

$$\phi\theta_1 k(I_1) - w_1 + \phi\theta_2 k(I_2) - w_2. \quad (12)$$

Interpretation: The CEO values output only through private benefits, scaled by ϕ , but cash wages reduce her utility one-for-one. This asymmetry makes capital an attractive compensation device.

5.5 Low, medium, and high productivity divisions

The model defines three productivity categories:

- θ_i is **low** if the manager rent-seeks even when he expects the efficient capital allocation;
- θ_i is **medium** if the manager's rent-seeking depends on whether the other manager rent-seeks;
- θ_i is **high** if the manager never rent-seeks because private benefits from staying are high enough.

Interpretation: These categories organize the model's five regions and determine when capital is distorted.

6 Main theoretical results

6.1 Proposition 1: regions of rent-seeking and capital allocation

Assume without loss of generality that $\theta_1 < \theta_2$. The model has five regions:

1. **Region 1:** θ_1 high or medium, θ_2 high. Neither manager rent-seeks; investment is efficient.
2. **Region 2:** θ_1 low, θ_2 high. Only the weak division manager rent-seeks; investment is distorted toward division 1.
3. **Region 3:** both divisions low. Both managers rent-see, but investment is efficient because both retention constraints bind and shifting capital does not save cash.
4. **Region 4:** θ_1 low, θ_2 medium. Both managers rent-see; investment may or may not be distorted toward the weak division.
5. **Region 5:** both divisions medium. There are two equilibria: a no-rent-seeking efficient equilibrium and a rent-seeking coordination-failure equilibrium.

Interpretation: The cleanest empirical prediction is Region 2: a weak division paired with a strong division receives inefficiently high capital.

6.2 Why the strongest division can subsidize the weakest division

In Region 2, the strong division manager has a slack retention constraint. The CEO can shift capital away from the strong division without needing to compensate its manager in cash. This makes capital reallocation a cheap way to pay the weak division manager.

Interpretation: Cross-subsidization is not driven by altruism or a desire to equalize outcomes. It is driven by CEO-level agency and the relative slackness of retention constraints.

6.3 Why distortions disappear when both divisions are weak

If both managers rent-seek, both retention constraints bind. Shifting capital from one division to another helps one manager but hurts the other. The CEO then has no cheap capital-allocation trick and uses cash wages instead.

Interpretation: Rent-seeking alone is not enough to generate capital misallocation. Misallocation requires asymmetry between divisions.

6.4 Proposition 2: magnitude of the investment distortion

When investment is distorted, the CEO shifts capital toward division 1 until one of three limits is reached:

1. the weak manager's retention constraint is satisfied with zero cash wage;
2. the marginal cost to the CEO of distorting investment becomes too high;
3. the strong manager's retention constraint starts to bind.

The marginal-cost condition can be written as

$$\phi[\theta_2 k'(I_2) - \theta_1 k'(I_1)] = \gamma \theta_1 k'(I_1). \quad (13)$$

Interpretation: The distortion is larger when the CEO has low-powered incentives, i.e., when ϕ is small. A low- ϕ CEO internalizes less of the lost output from inefficient capital allocation.

7 Investor choice of capital and operating budgets

7.1 Operating budget W

Outside investors may give the CEO an operating budget to pay wages. The paper shows that, in the baseline model, investors do not benefit by reducing W below the amount the CEO would spend on wages. Restricting cash wages would force the CEO to distort capital even more or fail to retain important managers.

Interpretation: The problem is not that the CEO pays too much cash. The problem is that she would rather conserve cash and pay with distorted capital allocations.

7.2 Capital budget I

If outside investors anticipate no misallocation, they set total investment equal to the efficient total. If they anticipate misallocation, they may reduce total investment.

Interpretation: The paper's main results are about the division of the capital budget across divisions, not the total size of the capital budget.

8 Extensions and variations

8.1 Signal-jamming interpretation

The CEO need not literally divert the operating budget. A similar logic arises if the CEO owns stock and wants to boost short-run reported profits. Holding down wages raises near-term earnings, and the CEO can substitute capital allocations for wage payments.

Interpretation: The model's mechanism does not require extreme CEO theft. It requires only that the CEO has some reason to economize on cash wages.

8.2 Breakups

The model predicts that inefficient conglomerates are natural breakup candidates. Breakups are most likely when divisional productivity is highly imbalanced and when the weak division receives excessive capital due to rent-seeking.

Interpretation: The theory links internal capital market inefficiency to the empirical wave of refocusing and divestitures.

8.3 Rent-seeking as signal-jamming or organizational politics

Rent-seeking may take many forms: external visibility, résumé polishing, entrenchment, manipulation of internal information, opacity in accounting systems, or political lobbying. The exact form is less important than its effect on bargaining power.

Interpretation: The model captures a broad class of internal influence activities that reduce productive effort and distort capital allocation.

9 Empirical implications

9.1 Prediction 1: weak divisions are subsidized when paired with strong divisions

The strongest prediction is that a weak division receives excessive investment when paired with a strong division. The strong division's manager has a slack retention constraint, making it easier for the CEO to reallocate capital away from that division.

Interpretation: This maps directly to empirical tests of whether low- q divisions receive more capital when other divisions in the same firm have high q .

9.2 Prediction 2: misallocation is strongest with high divisional imbalance

Capital distortions should be largest when productivity differences across divisions are large. If both divisions are weak or both are medium, the prediction is weaker or ambiguous.

Interpretation: This is why the theory predicts “socialism” most clearly when the firm contains both very strong and very weak divisions.

9.3 Prediction 3: CEOs with weaker incentives tolerate larger distortions

When the CEO's incentives to maximize shareholder value are weak, the marginal output loss from misallocation matters less to her. Therefore she will be more willing to satisfy division managers by distorting investment.

Interpretation: Stronger CEO incentives, better governance, or more active external monitoring should reduce inefficient cross-subsidization.

9.4 Prediction 4: breakups and refocusing reduce distortions

If a diversified firm breaks up, division managers become directly exposed to external capital markets and cannot as easily extract rents through internal capital allocations.

Interpretation: Breakups can improve investment efficiency by removing the internal political economy that allows weak divisions to extract subsidies.

10 Tables, figures, and original equation panels

10.1 Panel 1: model primitives and rent-seeking

Read:

- The model starts with two periods, two divisions, a CEO, outside investors, and division managers.
- Output depends on division productivity θ_i and managerial effort.
- Division managers and the CEO receive private benefits from output.
- Rent-seeking reduces productive effort but improves bargaining power.

Economic message: The primitive tension is between productive effort and influence activity. Low-productivity managers have lower opportunity cost of rent-seeking.

compensation.

B. Production and Rent-seeking When CEO Bargains With Cash Only

There are two periods, 1 and 2. At time 1, the CEO hires a new manager for each division, to work with assets that are already in place. The wages of these new managers at time 1 are normalized to zero. The output of division i at time 1 is $\theta_i f(e_i)$, where θ_i is a measure of the productivity of the assets in place, e_i is the productive effort of the division manager, and $f(\cdot)$ is an increasing concave function. The output is fully and costlessly verifiable, and hence can be assigned to outside investors. In addition to this verifiable output, there are also noncontractible private benefits. In particular, division manager i reaps a private benefit of $\gamma \theta_i f(e_i)$ from his own division, and the CEO gets a private benefit of $\phi \theta_i f(e_i)$ from each of the two divisions. These private benefits can be thought of in any number of ways: the usual perks, psychic benefits from empire building, and so forth. They are assumed to be small relative to the verifiable output—that is, γ and ϕ are both

10.2 Panel 2: cash-only benchmark

Read:

- The panel contains the retention constraint, the rent-seeking condition, and the first-order condition for rent-seeking effort.
- A low-productivity division manager is more likely to require a wage and more likely to rent-seek.
- In the cash-only benchmark, these wage demands do not distort investment because investment is fixed at the efficient allocation.

Economic message: The cash-only benchmark isolates the source of bargaining pressure before introducing capital-budget distortions.

Moreover, we assume that the CEO also has enough wealth to pay any cash wages. In this setting, the CEO always rehires the division manager at time 2, and faces the following “retention constraint”:

$$w_i + \gamma\theta_i k(I_i^*) \geq g(r_i). \quad (2)$$

In words, the CEO simply has to promise the division manager more total utility than he would get if he left. Note that the labor contract implied in this discussion is an easily implementable one. All that we require is that a court be able to verify (1) whether the division manager was in fact employed by the firm at time 2, and (2) whether the promised cash wage was paid. It is not necessary for us to assume that a court can verify capital expenditures at the divisional level. One can imagine that at time 2, the CEO first irreversibly sinks the capital allocations, and then, conditional on these allocations, promises to pay any still-needed cash wage to each division manager if he stays on.

The first observation about the retention constraint is that it may be slack at the efficient level of investment, I_i^* . If division i is very productive (i.e., has a high value of θ_i) and is therefore doing a lot of investing at the efficient level, the division manager gets more than his outside option in utility even without a cash wage, so the CEO can set $w_i = 0$. On the other hand, if the division has a low value of θ_i , it is more likely that the CEO has to pay a positive cash wage to keep the manager at time 2.

Now let us back up and consider the division manager's decision of how to divide his time between productive effort e_i and rent-seeking r_i . If the manager rent-seeks, his time 1 private benefits fall by $\gamma\theta_i[f(h) - f(h - r_i)]$ because of the reduced time 1 output. However, his time 2 outside option rises by $g(r_i)$. If $g(r_i)$ is greater than $\gamma\theta_i k(I_i^*)$ —so that the retention constraint is violated when the wage is zero—then the CEO must pay a wage of $g(r_i) - \gamma\theta_i k(I_i^*)$ to keep the manager. Thus, for a given r_i the manager rent-seeks, provided the time 2 gain in wage compensation exceeds the loss in time 1 private benefits:

$$g(r_i) > \gamma\theta_i[k(I_i^*) + f(h) - f(h - r_i)]. \quad (3)$$

If there is an r_i that satisfies this “rent-seeking condition,” we know that there is at least some rent-seeking in equilibrium. Just how much, r_i^* , is determined by equating the marginal gain in time 2 compensation and the marginal loss in time 1 private benefits:

$$g'(r_i^*) = \gamma\theta_i f'(h - r_i^*). \quad (4)$$

Note, however, that if at r_i^* the rent-seeking condition

$$g(r_i^*) > \gamma\theta_i[k(I_i^*) + f(h) - f(h - r_i^*)] \quad (5)$$

10.3 Panel 3: low, medium, and high productivity thresholds

Read:

- Productivity types are defined by whether rent-seeking is always attractive, conditionally attractive, or never attractive.
- The low/medium/high classification determines the five regions of Proposition 1.
- Strategic complementarity arises because one manager's rent-seeking can reduce the other manager's expected capital allocation.

Economic message: The low/medium/high definitions translate the informal idea of organizational politics into precise conditions.

written as $\gamma\theta_i I_i(r_j)$, where $I_i(r_j)$ denotes the investment allocation to i if i does not rent-peek and the other manager, j , rent-seeks at level r_j . This implies that manager i rent-seeks if

$$g(r_i^*) > \gamma\theta_i [k(I_i(r_j)) + \Delta f_i]. \quad (9)$$

As will become clear, the capital allocation to i tends to be reduced below the efficient level if only j actively rent-seeks: $I_i(r_j^*) < I_i^*$. It follows that i 's incentive to rent-peek can be greater if j rent-seeks—this is the sense in which rent-seeking activities can be strategic complements. Note that this is the case even though manager j 's rent-seeking does not directly reduce the rents available to manager i .

Now let us make the following further definitions. We say that

$$\theta_i \text{ is "low" if } g(r_i^*) > \gamma\theta_i [k(I_i^*) + \Delta f_i] \quad (10)$$

$$\theta_i \text{ is "medium" if } \gamma\theta_i [k(I_i^*) + \Delta f_i] \geq g(r_i^*) > \gamma\theta_i [k(I_i(r_j^*)) + \Delta f_i] \quad (11)$$

$$\theta_i \text{ is "high" if } g(r_i^*) \leq \gamma\theta_i [k(I_i(r_j^*)) + \Delta f_i]. \quad (12)$$

The interpretation of these definitions is as follows. When θ_i is low, manager i wishes to rent-peek no matter what—even if he conjectures that, absent rent-seeking, he will be allocated the efficient level of investment I_i^* . In contrast, when θ_i is high, manager i never rent-seeks, even if he conjectures that, absent rent-seeking, the capital budget will be tilted to manager j , so that he (manager i) gets a reduced allocation of only $I_i(r_j^*)$. Finally, when θ_i is medium, we have a case where there is the potential for strategic interaction: manager i 's rent-seeking decision *depends* on his conjecture about how much capital he gets if he does not rent-peek.

With the definitions in hand, we can now provide a characterization of the solution to the CEO's problem. The full details are in Appendix A. Here we just state the main qualitative results.

PROPOSITION 1: *Assume without loss of generality that $\theta_1 < \theta_2$. There are five regions to be considered:*

10.4 Panel 4: Proposition 1

Read:

- Proposition 1 classifies the equilibrium according to the productivity of the two divisions.
- Region 2 is the main region: weak division 1 rent-seeks, strong division 2 does not, and investment is tilted toward division 1.
- Region 3 shows that rent-seeking alone does not imply misallocation if both divisions are weak.
- Region 5 shows a possible coordination failure in which both medium-productivity managers rent-peek.

Economic message: Capital socialism is most likely when a weak division is paired with a strong division, not when all divisions are weak.

PROPOSITION 1. Assume without loss of generality that $\theta_1 < \theta_2$. There are five regions to be considered:

- (1) *Region 1: θ_1 high or medium; θ_2 high. In this region, neither division manager rent-seeks, and investment is efficient: $I_1 = I_1^*$.*
- (2) *Region 2: θ_1 low; θ_2 high. In this region, only the manager of division 1 rent-seeks, and investment is distorted towards division 1: $I_1 > I_1^*$.*
- (3) *Region 3: θ_1 low; θ_2 low. In this region, both division managers rent-peek. However, investment is efficient: $I_1 = I_1^*$. The retention constraints of both managers are met with cash wages: $w_i > 0$.*
- (4) *Region 4: θ_1 low; θ_2 medium. In this region, both division managers rent-peek. Depending on parameter values, investment may or may not be distorted towards division 1: $I_1 \geq I_1^*$.*

Proposition 1, first regions

Divisional Rent-Seeking and Inefficient Investment 2551

- (5) *Region 5: θ_1 medium; θ_2 medium. In this region, there are two possible equilibria, which we denote by Equilibrium 5a and Equilibrium 5b. Equilibrium 5a: Neither division manager rent-seeks, and investment is efficient: $I_1 = I_1^*$. Equilibrium 5b: Both division managers rent-peek. Depending on parameter values, investment may or may not be distorted towards division 1: $I_1 \geq I_1^*$.*

Region 1, where both divisions are relatively high productivity, represents the simplest and least interesting case. In this portion of the parameter

Proposition 1, continuation

10.5 Panel 5: Proposition 2

Read:

- Proposition 2 gives the stopping rules for the CEO's distortion of the capital budget.
- The CEO keeps shifting capital to the weak division until retention is satisfied, the marginal distortion cost becomes too large, or the strong manager's retention constraint binds.
- The distortion is amplified when the CEO's incentive parameter ϕ is small.

Economic message: Proposition 2 explains why internal capital market distortions can be economically large under weak CEO incentives.

capital to division 1 until one of the following limits is hit:

- (1) Division manager 1's retention constraint is satisfied with a zero cash wage, i.e.: I_1 solves $g(r_1^*) = \gamma\theta_1 k(I_1)$; or
- (2) The marginal cost to the CEO of distorting investment becomes so high that she is unwilling to distort it further, and prefers to pay the rest of the needed compensation to the manager of division 1 in cash: $\phi(\theta_2 k'(I_2) - \theta_1 k'(I_1)) = \gamma\theta_1 k'(I_1)$; or
- (3) The retention constraint of division manager 2 binds when $w_2 = 0$: $g(r_2^*) = \gamma\theta_2 k(I_2)$.

The implication of Proposition 2 is that we are likely to get a large distortion in investment allocation when the following conditions jointly obtain: (1) division 1 is very weak, so that θ_1 is very small and a large value of I_1 is needed to satisfy manager 1's retention constraint; (2) division 2 is very strong, so that manager 2 never rent-seeks and always has a slack retention constraint; and (3) the CEO has low-powered incentives (i.e., ϕ is small) so that she is willing to tolerate a large difference in the marginal product of capital across the two divisions.

To isolate the pure effects of agency at the CEO level, note that from part 2 of Proposition 2, an upper bound on the investment distortion is given by $\theta_2 k'(I_2)/\theta_1 k'(I_1) = 1 + \gamma/\phi$. If ϕ and γ are of the same order of magnitude,

²¹ When one division manager's retention constraint is slack, and the principal initially shifts

10.6 Panel 6: signal-jamming extension

Read:

- The model can be restated without assuming literal diversion of the operating budget.
- If the CEO wants to boost reported profits or short-run stock price, she may still prefer conserving cash wages.
- Capital can then substitute for cash compensation in divisional bargaining.

Economic message: The mechanism is robust: the CEO only needs a reason to value cash conservation more than efficient divisional capital allocation.

in these figures as the parameter space varies may, the manager's communications, $\bar{I}$ is set at a lower value.

II. Extensions and Variations

A. A Signal-jamming Rationale for Conserving the Operating Budget

Up to now, we have assumed that the auditing/verification technology is such that the CEO has some discretion over the operating budget—if the CEO does not spend the operating budget on division-manager wages, she can instead spend it on something else that raises her utility, such as excess travel. In contrast, the CEO has less discretion with respect to the capital budget—although she can choose in which division to spend it, the funds must be spent on physical capital somewhere in the firm, and cannot be used for other purposes. These assumptions taken together lead to our principal conclusion, namely that the CEO may prefer to tilt the capital budget rather than spend the operating budget to satisfy a division manager's demands.

Thus a natural criticism of our model is that our results are a by-product of an implausible assumption. Why, it might be asked, can't outside investors somehow circumscribe the CEO's ability to divert the operating budget? For example, with a stronger auditing technology, investors might be able to specify that the funds they put up for division-manager wages cannot be used for anything else. This would prevent the CEO from enjoying private benefits when she holds down division-manager wages.

In response to this critique, we should stress that it is not really necessary for our results that the CEO be able to divert the operating budget to herself. All that is really required is the much weaker condition that the CEO have some reason to care about economizing on wage payments to division managers. We now sketch one alternative model—without any diversion of the operating budget—which has this property.

The alternative model is a variation on Stein (1989). Assume that the CEO owns stock in the firm, and that she contemplates selling some of this stock in the not-too-distant future. Thus the CEO wishes to take actions that increase the short-run stock price, even if these actions ultimately reduce the

10.7 Panel 7: empirical implications

Read:

- The empirical section connects the model to evidence on conglomerate investment, segment-level investment, and divestitures.
- The theory predicts overinvestment in weaker divisions and sharper distortions when productivity differences across divisions are large.
- The model also predicts that breakups can improve investment efficiency.

Economic message: The theory gives a microfoundation for empirical evidence on conglomerate discounts and internal capital market socialism.

Rajan et al. (2000) use a similar data set, and uncover three noteworthy patterns. First, consistent with the notion of socialism, multisegment firms allocate relatively more than their stand-alone counterparts to segments in “weak” lines of business (as measured either by industry Q ratios or investment rates) and relatively less to segments in “strong” lines of business. Second, this misallocation of resources is most pronounced when there is a wide disparity in the productivity of investment across the lines of business, as proxied for by the dispersion of the corresponding industry Q ratios. This is directly in line with the predictions of our Propositions 1 and 2, which state that large investment inefficiencies are most likely to occur when there is a pronounced differential in divisions’ investment opportunities. Finally, the investment distortions have significant value consequences—those diversified firms that have the most heterogeneous segments and that misallocate funds the most also tend to trade at the largest discounts.²⁵

Scharfstein (1998) also finds evidence of socialism in a sample of diversified conglomerates, in that the investment of conglomerate divisions is virtually insensitive to their investment opportunities, as measured by the corresponding industry Qs. (In contrast, the investment of stand-alone firms is significantly more sensitive to industry Qs.) But perhaps most interesting from the perspective of our model is Scharfstein’s finding that socialism seems to be driven by a misalignment of incentives between outside investors and top management. In particular, he shows that the sensitivity of divisional investment to Q increases as top management’s equity stake in the firm goes up. Or, said differently, socialism is more pronounced when the CEO has low-powered incentives. This gets precisely at the heart of the most distinctive aspect of our model—the idea that the agency problem between the CEO and outside investors is a crucial part of the story. As noted earlier, the model of Rajan et al. (2000), where investment levels are set by a principal rather than by an agent, does not make the same prediction.

Some of the evidence put forth by Shin and Stulz (1998), Rajan et al. (2000), and Scharfstein (1998) has been challenged in a recent paper by Chevalier (1999), who argues that there may be biases arising from unobserved differences between conglomerate divisions and stand-alone firms. However, Gertner, Powers, and Scharfstein (1999) demonstrate that even when one controls for these unobserved differences—by examining the same division before and after it is spun out of a conglomerate—investment is markedly less sensitive to Q when a division is inside a conglomerate.

Thus overall, evidence is beginning to emerge that suggests that not only is there a general tendency towards socialism in internal capital markets, but that this problem is more acute when (1) there are wide disparities across divisions in investment prospects, and (2) top management has weak incentives to maximize value. This broad picture squares very well with our theory.

²⁵ Berger and Ofek (1995) also find that the conglomerate discount is greatest in firms that

11 Short summary

- The paper studies how divisional rent-seeking can distort internal capital markets.
- Division managers can spend effort improving their bargaining position instead of producing output.
- Low-productivity managers are more likely to rent-seek because productive effort is less privately valuable to them.
- The CEO must retain division managers but is herself an agent of outside investors.
- Because the CEO wants to conserve cash wages, she may compensate rent-seeking managers with capital allocations.
- This creates inefficient cross-subsidization, especially when a weak division is paired with a strong division.
- When both divisions are weak, investment may remain efficient because both retention constraints bind.

- When both divisions are medium productivity, the firm may have multiple equilibria, including a dysfunctional rent-seeking equilibrium.
- Distortions are larger when CEO incentives are weaker.
- The model explains why diversified firms may overinvest in weak divisions and why breakups may improve efficiency.

12 Conclusion

12.1 Core conclusion

Scharfstein and Stein show that internal capital markets can have a dark side. Headquarters authority over capital allocation does not guarantee efficient reallocation. When division managers can rent-seek and CEOs care about conserving cash, capital budgets become a form of implicit compensation. This can produce inefficient subsidies to weak divisions.

12.2 Economic intuition

A low-productivity division manager has relatively little to lose from diverting effort into rent-seeking. If he becomes costly to lose, the CEO must compensate him. Paying cash wages is privately expensive for the CEO, but shifting capital toward the manager's division initially looks cheaper, especially if the strong division manager has a slack retention constraint. The CEO therefore chooses a distorted capital allocation even though outside investors would prefer efficient investment.

12.3 Incremental contribution takeaway

The paper's incremental contribution is a two-tier agency theory of internal capital market socialism. It explains why diversified firms may not simply allocate capital poorly because CEOs like empire building, but because divisional politics and CEO-level agency interact. This produces a precise prediction: weak divisions are most likely to be subsidized when paired with strong divisions.

12.4 Final takeaway

Internal capital markets are not automatically superior to external capital markets. Their efficiency depends on the internal political economy of the firm: who bargains, who controls budgets, and whether cash or capital is used to settle internal claims.